Paidala Vindhya

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.38
Narayana Junior College, Tirupati	Class XII (BIEAP (Board of Intermediate Education, Andhra Pradesh))	2023	98.70%
Narayana E.M High School, Tirupati	Class X (BSEAP (Board of Secondary Education, Andhra Pradesh))	2021	99.67%

EXPERIENCE

Microsoft | SWE Intern | Linux App Service

- **Oryx Decoupling** | C#, Docker, Linux, Azure DevOps
- Decoupled the Oryx build CLI from Kudu deployment container into a minimal **150MB OS-agnostic image**.
- Optimized fleet storage by removing OS-specific Oryx images, saving ~300MB per VM and **~300TB** across the fleet.
- **Eliminated 3GB Kudu image rebuilds** for every Oryx update and enabled independent release trains.
- Engineered dynamic volume mounting logic in worker VMs governed by feature flags for instant rollbacks, validating end-to-end deployments by triggering builds of various app stacks.
- **CVE Agent** | Azure Logic Apps, GPT-4o mini, GitHub Actions, Regctl, Trivy, AAD
- Developed a **Teams-triggered CVE system** using Azure Logic Apps and GPT-4o mini for natural language **intent detection**.
- Automated Trivy vulnerability scans across **14 release stages** (Stages 0-6, standard/latest) using GitHub Actions.
- Secured workflow callbacks via AAD authentication and optimized execution time by caching image tag results.

PROJECTS

Concurrent Merkle Tree Operations | Research | Prof. Sathya Peri

- Developed a **linearizable** concurrent Merkle tree update algorithm via fine-grained, **hand-over-hand locking** in a team of 2.
- Implemented a branch snapshot mechanism to provide consistent views during concurrent execution.
- Achieved **higher throughput** and **lower response time** than sequential and Angela baselines.

RISC-V Simulator | Prof. Rajesh Kedia

- Developed a RISC-V ISA simulator supporting assembly parsing, instruction decoding, and execution.
- Implemented breakpoints, step/run execution, memory/register inspection, stack viewing, and cache simulation.

Virtual Memory Management in GemOS | Prof. Saurabh Kumar

- Implemented mmap, munmap, and mprotect **system calls** to manage VMAs with lazy allocation and 4-level page tables.
- Implemented cfork with Copy-on-Write for efficient memory sharing and correctly handled CoW page faults.

Concurrent Synchronisation & Data Structures | Prof. Sathya Peri

- Built lock-free queues and Multi-Reader Single-Writer regular registers for safe thread access.
- Implemented TAS, TTAS, CLH, and MCS spin locks to compare concurrent list operations performance.
- Implemented static and dynamic thread allocation methods to balance parallel workloads and reduce overhead.

Finex Compiler | Prof. Ramakrishna Upadrasta

- Built a statically-typed **domain-specific language** compiler for quantitative finance in a team of 6.
- Built a Flex-based lexer and Bison LALR(1) parser supporting custom financial tokens to generate an AST class hierarchy.
- Implemented a 2-pass visitor-based semantic analyzer with nested scopes, symbol tables, and type-checking.

Secure File Management Server | Prof. Saurabh Kumar

- Built a multithreaded C++ client-server architecture driven by a custom **TLV protocol** over TCP.
- Implemented nonce **authentication** and **Diffie-Hellman key exchange** to establish secure encrypted sessions.
- Integrated AES-128 E2EE in CBC mode and SHA-256 MACs for chunk-based file integrity verification.

Dashboard Admin Console | IITH Lambda Club

- Collaborated on the Dashboard website as part of **Lambda**, developing the faculty room-booking frontend in **React**.
- Built the club event room-booking app using **Flutter** & managed user data via **Dio APIs**.

POSITIONS OF RESPONSIBILITY

- **Core** at **Lambda**, The Software Development Club of Sci-Tech Council, IITH.
- **Teaching Assistant** for **Operating Systems** and **Database Management Systems.**
- Deputy Coordinator for Techy Events at **Elan & nVision 2024**, the Cultural and Technical Fest of IITH.

SKILLS

- **Languages** : C, C++, C#, SQL, Bash, Python, RISC V Assembly
- **Web/App Development :** HTML, CSS, JavaScript, React.js, ShadCN, TailWind CSS, Node.js, Express.js, Dart, Flutter
- **DevOps & Tools :** Azure App Service, Azure Logic Apps, Azure DevOps, Docker, Linux, GitHub Actions, YAML, Trivy, Git, GitHub, MySQL, PostgreSQL, MongoDB, VS Code, LaTeX, Lex, MakeFile, Google App Script

RELEVANT COURSES

- **Computer Science** : Parallel & Concurrent Programming, Network Security, Operating Systems, Database Management Systems, Compilers, Computer Networks, Algorithms, Data Structures, Computer Architecture, Software Engineering, Theory of Computation, Foundations of Machine Learning, Deep Learning, Discrete Maths
- **Mathematics** : Probability & Random variables, Transform Techniques, Linear Algebra, Calculus, Statistics, Linear Optimization

EXTRACURRICULAR

- Member of **National Service Scheme(NSS)**, IITH.
- Volunteered in the **Aksharmalaa** online teaching program for rural government school students.